CLASS 9 – ALGEBRA

Problems:

1. Explore the Factor Theorem for $p(x) = x^3 - 4x^2 + x + 6$

2. Solve a pair of linear equations using cross multiplication

3. Explore consistency of a pair of linear equations

4. Factorise the quadratic polynomial $x^2 + 5x + 6$

5. Explore the Remainder Theorem using a slider

Geogebra Commands:

1. Explore the Factor Theorem for $p(x) = x^3 - 4x^2 + x + 6$

This activity checks whether $(x - a)$ is a factor by changing a and observing p(a).

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Enter the polynomial in the Input Bar:

p(x) = x^3 - 4x^2 + x + 6

Step 3: Create a slider for a:

a = Slider(-2, 4, 1). Set the initial value of a to 1 if needed.

Step 4: Calculate p(a):

v = p(a)

Step 5: Create a point on the curve:

A = (a, p(a))

Step 6: Display the polynomial:

t1 = Text("Polynomial: p(x) = x^3 - 4x^2 + x + 6", (-4, 8))

Step 7: Display the current value:

t2 = Text("Current value p(a) = " + v, (-4, 7))

Step 8: Display the theorem:

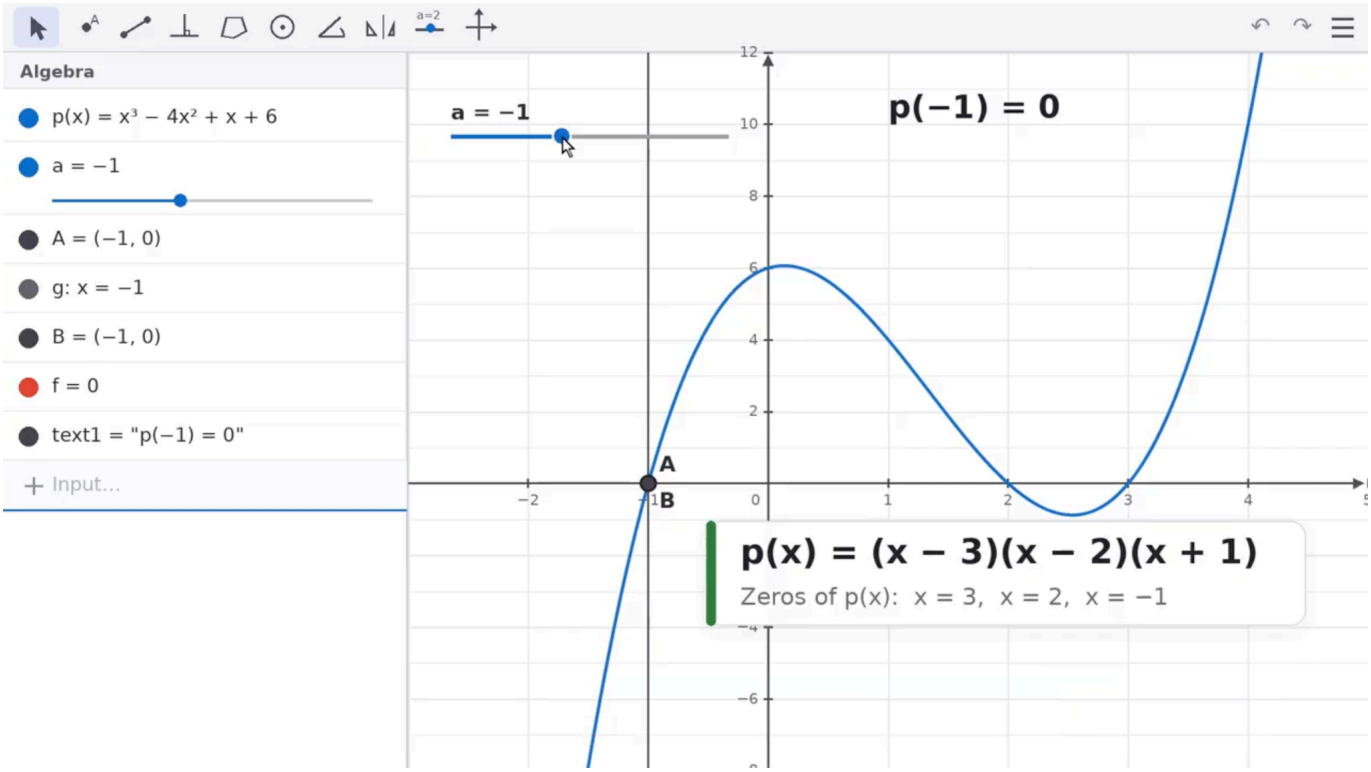
t3 = Text("If p(a) = 0, then (x - a) is a factor.", (-4, 6))

Step 9: Move the slider and observe:

a = 1 gives p(1) = 4; a = −1 gives p(−1) = 0; a = 2 gives p(2) = 0; a = 3 gives p(3) = 0.

Output:

The graph crosses the x-axis at −1, 2, and 3. Therefore, $(x + 1)$, $(x - 2)$, and $(x - 3)$ are factors. Final factorisation: $p(x) = (x + 1)(x - 2)(x - 3)$.



2. Solve a pair of linear equations using cross multiplication

Question: Solve 2x + 3y = 13 and x − y = 1.

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Enter the first equation:

l: 2x + 3y = 13

Step 3: Enter the second equation:

m: x - y = 1

Step 4: Find the intersection point:

P = Intersect(l, m)

Step 5: Create the coefficients in standard form:

a1 = 2

b1 = 3

c1 = -13

a2 = 1

b2 = -1

c2 = -1

Step 6: Calculate the cross-multiplication values:

Nx = b1*c2 - b2*c1

Ny = c1*a2 - c2*a1

D = a1*b2 - a2*b1

Step 7: Calculate x and y, provided D ≠ 0:

xSol = Nx / D

ySol = Ny / D

Step 8: Display the solution:

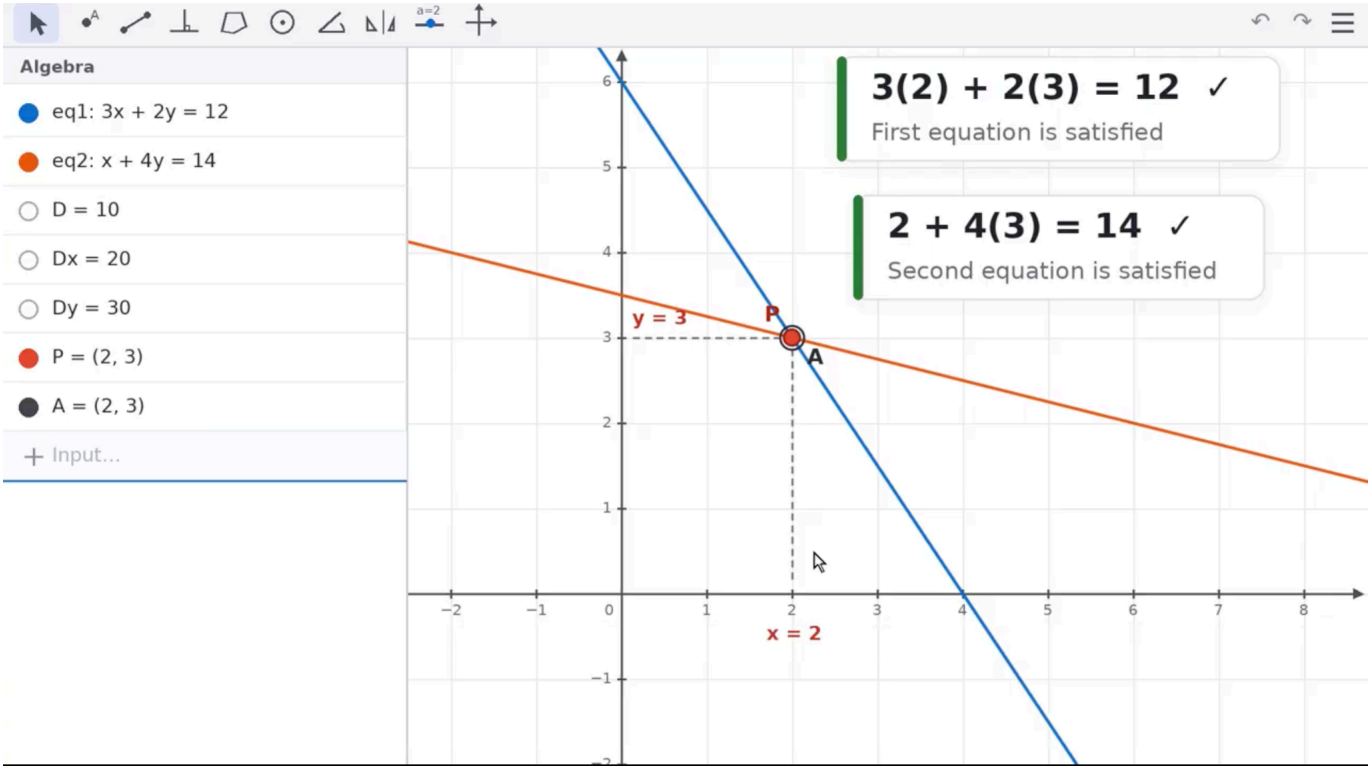
t4 = Text("Solution: x = " + xSol + ", y = " + ySol, (-4, 8))

Step 9: Verify the answer.

x = 16/5 = 3.2 and y = 11/5 = 2.2. Substitution gives 2x + 3y = 13 and x − y = 1.

Output:

GeoGebra shows the two lines intersecting at P = (3.2, 2.2). If D = 0, do not divide by zero; check whether the lines are parallel or coincident.



3. Explore consistency of a pair of linear equations

This activity demonstrates the three cases: one solution, no solution, and infinitely many solutions.

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Case 1 – Intersecting lines. Enter:

l: x + y = 5

m: x - y = 1

P = Intersect(l, m)

The lines meet at P = (3, 2). There is one solution; the pair is consistent and independent.

Step 3: Case 2 – Parallel distinct lines. Enter:

p: x + y = 4

q: x + y = 7

The lines never meet. There is no common solution; the pair is inconsistent.

Step 4: Case 3 – Coincident lines. Enter:

r: x + y = 4

s: 2x + 2y = 8

The equations draw the same line. Every point on the line is a common solution.

Step 5: Display the three conclusions:

t5 = Text("Intersecting: one solution", (-4, 8))

t6 = Text("Parallel distinct: no solution", (-4, 7))

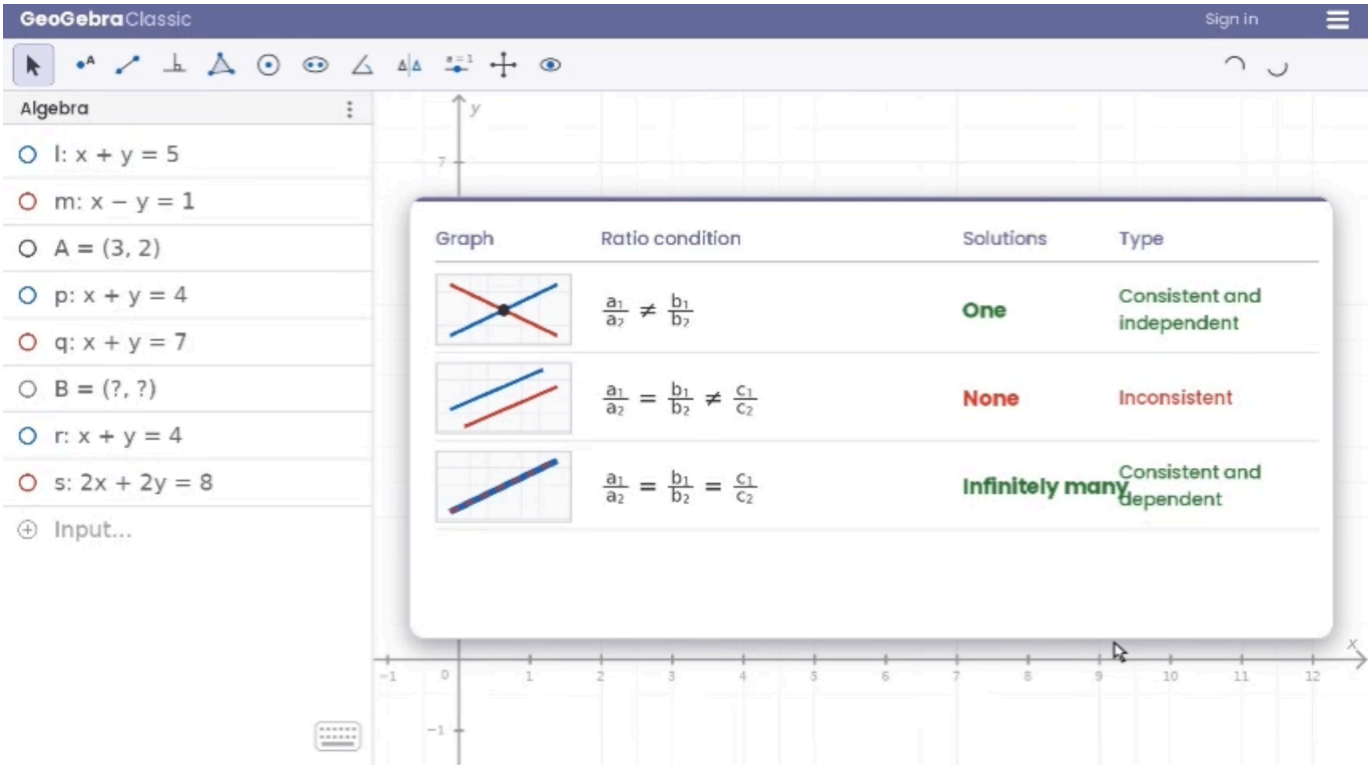
t7 = Text("Coincident: infinitely many solutions", (-4, 6))

Step 6: Check the coefficient-ratio conditions.

Intersecting: $a_1/a_2 \neq b_1/b_2$. Parallel distinct: $a_1/a_2 = b_1/b_2 \neq c_1/c_2$. Coincident: all three ratios are equal.

Output:

Intersecting lines have one solution; parallel distinct lines have no solution; coincident lines have infinitely many solutions. Write equations in standard form before applying the ratio test.



4. Factorise the quadratic polynomial $x^2 + 5x + 6$

This activity uses the product-and-sum method and the graph to verify the factorisation.

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Enter the original polynomial:

f(x) = x^2 + 5x + 6

Step 3: Enter the factorised expression:

g(x) = (x + 2)*(x + 3)

Step 4: Mark the roots:

A = (-3, 0)

B = (-2, 0)

Step 5: Display the original polynomial:

t8 = Text("Original: x^2 + 5x + 6", (-5, 8))

Step 6: Display the factorisation:

t9 = Text("Factors: (x + 2)(x + 3)", (-5, 7))

Step 7: Display the expansion check:

t10 = Text("Expansion: x^2 + 3x + 2x + 6", (-5, 6))

Step 8: Use the product-and-sum method.

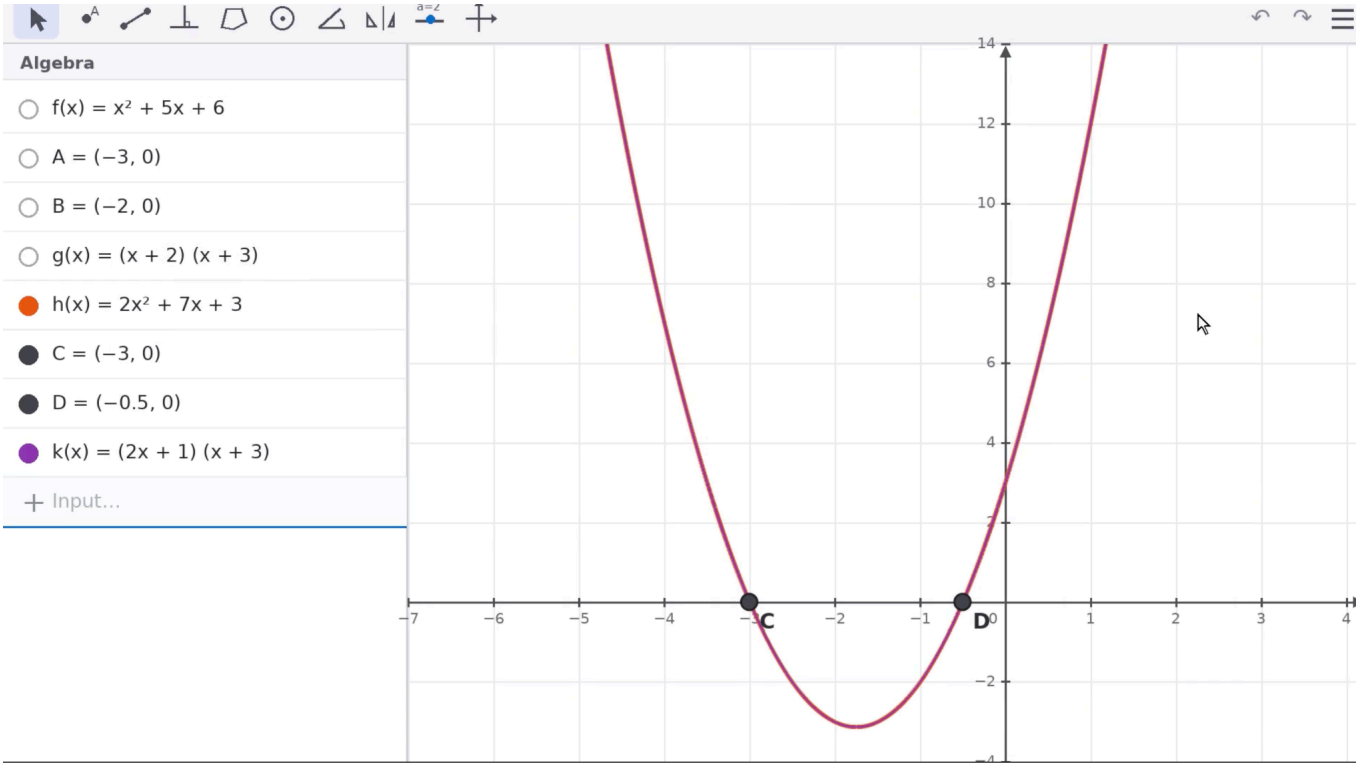
Find two numbers whose product is 6 and whose sum is 5. The numbers are 2 and 3.

Step 9: Observe the graph.

The graphs of f and g overlap because the expressions are equal. The x-intercepts are −3 and −2.

Output:

$x^2 + 5x + 6 = (x + 2)(x + 3)$. The roots are $x = -2$ and $x = -3$. The expression $2x^2 + 7x + 3$ is not equivalent to the original polynomial.



5. Explore the Remainder Theorem using a slider

This activity demonstrates how to evaluate a polynomial at a value a. When a polynomial p(x) is divided by (x − a), the remainder is p(a).

Question: Explore $p(x) = x^3 - 3x^2 + 4x + 50$ for different values of a.

Step 1: Open GeoGebra Classic and select the Graphing view.

Step 2: Enter the polynomial in the Input Bar:

p(x) = x^3 - 3x^2 + 4x + 50

Step 3: Create a slider for a:

a = Slider(-3, 5, 0.1)

Set the initial value of a to 0. If GeoGebra creates a number rather than a slider, enable Show Slider from the object's menu.

Step 4: Create a point on the polynomial graph:

A = (a, p(a))

Step 5: Display the current value p(a):

t1 = Text("Remainder for divisor (x - a) = " + p(a), (-4, 8))

Step 6: Display the current value of a:

t2 = Text("Current value of a = " + a, (-4, 7))

Step 7: Move the slider and observe the point A and the value p(a).

For example, when a = 0, p(0) = 50. The point A is then (0, 50).

Output:

The value p(a) gives the remainder when p(x) is divided by (x − a). The slider changes a, so the point and displayed value update automatically.

Important note: this is the Remainder Theorem for a linear divisor of the form x − a.

